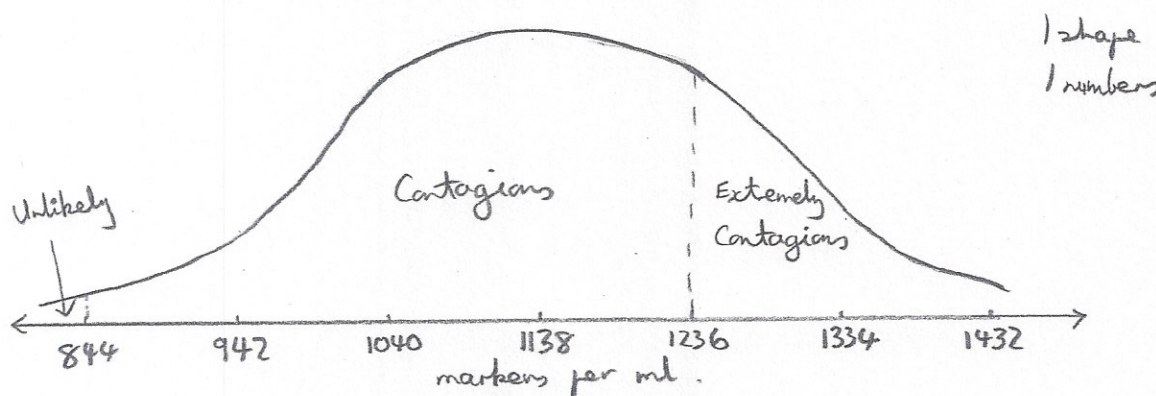


PART TWO (41 marks)NAME: SOLUTIONS TEACHER: JGA**Categorising infectiousness.**

From previous research, Professor Higgins has found that the *markers* in adult rats with Corona virus, is normally distributed with a mean of 1138 markers and a standard deviation of 98 markers.

1. Draw the bell-shaped curve representing the normal distribution with a mean of 1138 and a standard deviation of 98. Include values three standard deviations from the mean. (2 marks)



Professor Higgins has developed a classification system depending on the *markers* in the rat. The classification system is based on the following:

2. Complete the following table. (2 marks)

Classification	Interpretation	
Extremely contagious	Above one standard deviation above the mean	$markers \geq 1236$
Contagious	Between 3 standard deviations below the mean and 1 standard deviation above the mean.	$844 \leq markers < 1236$
Unlikely to be contagious	Below three standard deviations below the mean	$markers < 844$

3. Approximately how many of the 1000 rats fall into the extremely contagious category? (1 mark)

$$\frac{16}{100} \times 1000 = 160$$

4. Professor Higgins took a sample of 4 rats and calculated their standardised score.

Rat	z-score	Marker classification
A	-3.4	Unlikely
B	1.3	Extremely
C	-0.8	Contagious
D	3.1	Extremely

- a. State the marker classification for each rat by completing the above table. (2 marks)

- b. Confirm your classification in question 4a by calculating the *markers* for Rat A, Rat B, Rat C & Rat D. (2 marks)

$$\text{RAT A} \quad -3.4 = \frac{x - 1138}{98}$$

$$\text{RAT C} \quad -0.8 = \frac{x - 1138}{98}$$

$$x = 804.8$$

$$x = 1059.6$$

Unlikely

Contagious

$$\text{RAT B} \quad 1.3 = \frac{x - 1138}{98}$$

$$\text{RAT D} \quad 3.1 = \frac{x - 1138}{98}$$

$$x = 1265.4$$

$$x = 1441.8$$

Extremely

Extremely

Here is a reminder from PART ONE to help you answer the following questions.

Markers (per ml)	600 - < 700	700 - < 800	800 - < 900	900 - < 1000	≥ 1000
High Dosage	8	15	58	68	72

This is the five number summary for the 'High Dosage' group. Note that there are no outliers for this group.

Min=623, $Q_1=784$, Med=946, $Q_3=1076$, Max=1325

5. Using the information above:

- a. Approximately what percentage of the 'High Dosage' group are expected to have less markers than 784? (1 mark)

25%

- b. Approximately how many rats from this group are expected to be classified as 'Unlikely to be Contagious'? (2 marks)

25% of 221 = 55.25
 ≈ 55 rats

- c. How does this answer to 5b above compare to the normal distribution in question 2? (2 marks)

This 25% is far higher than the 0.15% expected in the normal distribution.

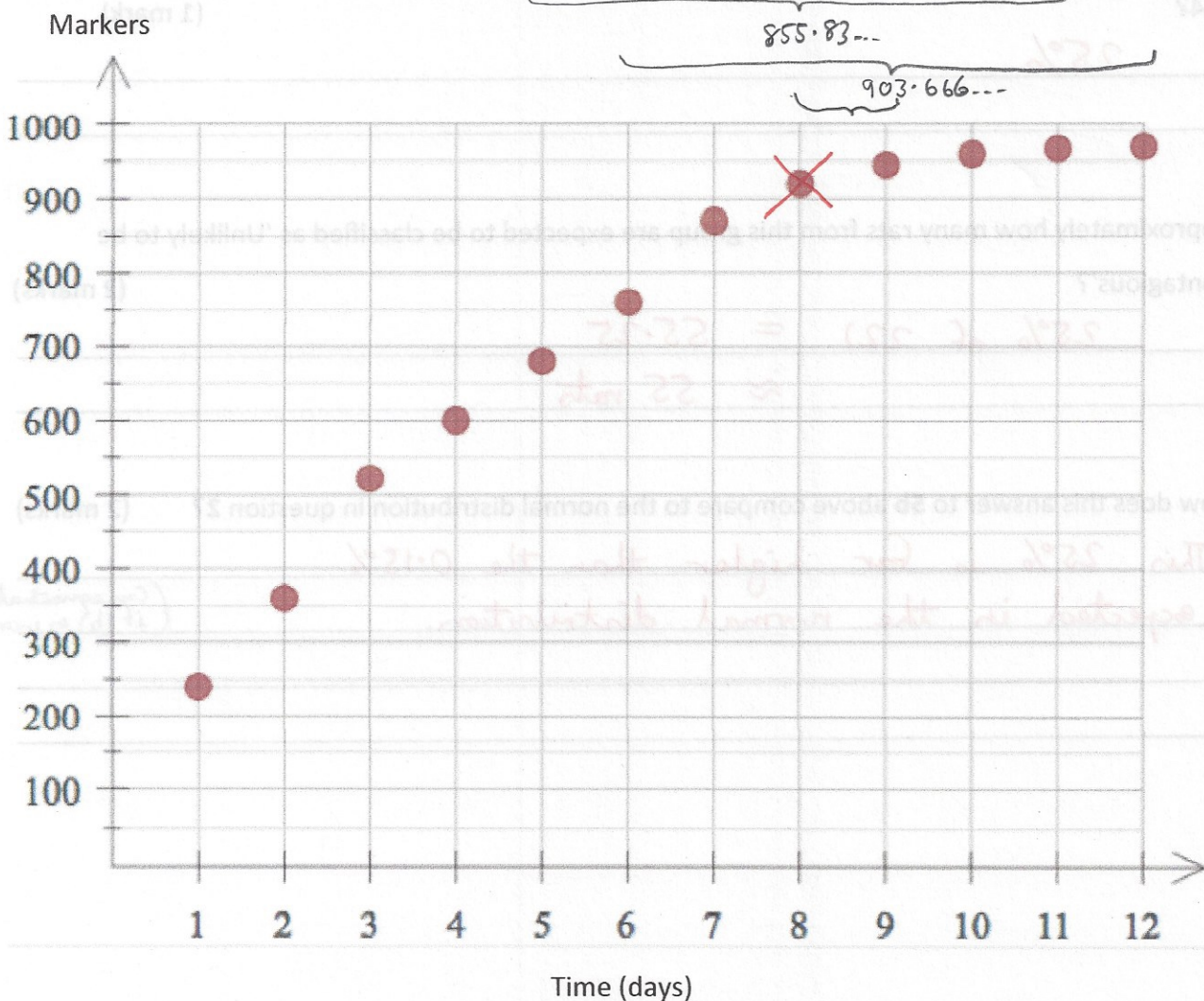
(Consequential - if (b) is incorrect)

Finding a model.

Professor Higgins has realised that many of the rats given a high dosage of the drug have more markers than the average markers of an infected adult rat. From previous experience, she has noticed that the internal organs of the rat stop working effectively once the rats reach a certain level of the markers in the bloodstream. Professor Higgins needs to develop a model to predict whether the rats given the high dosage will continue gaining markers.

The following table and scatterplot shows the average markers of the rats in the 'High Dosage' group over the period of 12 days.

Time (days)	1	2	3	4	5	6	7	8	9	10	11	12
Markers per ml	240	360	522	600	680	760	870	920	945	960	967	970



6. What is the form of this scatterplot?

(1 mark)

Non-linear.

7. What would be the problem with fitting a linear regression model to this data? (1 mark)

It wouldn't be a real representation of how the markers increase and therefore you would get incorrect/unreliable (high?) values if you were to predict markers using a linear model.

Correlation coefficient will not be accurate to measure the strength of the linear association.

8. Which transformations should be considered in order to linearise this data? (1 mark)

y^2
 $\log(x)$
 $\frac{1}{x}$

9. Complete the following table using the transformations you chose in question 8. (3 marks)

Transformation	Coefficient of determination (to 2 decimal places)	Residual plot
y^2	0.95	Residual values not close to zero (-120 000 to 120 000). Not random - Pattern. (1)
$\log(x)$	0.98	Residual values are closer to zero (-60 to 60). Random? Pattern? (1)
$\frac{1}{x}$	0.80	Residual values further from zero (-120 to 120). Not random - Pattern. (1)

10. Which transformation is the best? Explain why. (2 marks)

The best transformation is the $\log(x)$ transformation as it has the highest r^2 value (0.98) the residual values are most randomly distributed and the residuals are closest to zero. (2 reasons)

11. What is the equation of this transformation model to 3 significant figures and using appropriate variables? (2 marks)

$$\text{markers} = 180 + 764 \log(\text{days})$$

12. How many markers are the 'High Dosage' rats expected to have after they have been infected for three weeks to the nearest number? (2 marks)

Sub 21 days.

$$\begin{aligned} \text{markers} &= 180 + 764 \log(21) \\ &= 1190.1755... \\ &\approx 1190 \text{ markers} \end{aligned}$$

13. The internal organs of a rat will stop working approximately when the average number of markers is more than 1400.

- a. When is it predicted that the 'High Dosage' rats will reach this level to 2 significant figures? (2 marks)

$$\text{Solve } (1400 = 180 + 764 \log(\text{days}), \text{ days})$$

$$\begin{aligned} \text{days} &= 39.5237... \\ &\approx 40 \text{ days} \end{aligned}$$

- b. What are the limitations of this prediction? (1 mark)

Unreliable as prediction is outside the data range and is extrapolation.

Making predictions and the current effect on the world.

Using the High Dosage data on page 16, the transformation equation for is found to be:

$$\text{markers} = 180 + 764(\log(\text{days}))$$

14. What is the residual for day 8 to 2 decimal places?

(1 mark)

$$\text{Residual} = \text{Actual} - \text{Predicted}$$

$$\text{Predicted} = 180 + 764 \log(8) = 869.96 \dots$$

$$\text{Actual} = 920$$

$$\text{Residual} = 920 - 869.96 \dots = 50.04$$

15. According to the classification system, rats are considered 'Contagious' if their markers are more than 884.

a. After how many days to 1 decimal place, would you expect the 'High dosage' rats to be considered contagious?

(2 marks)

Use Solve.

$$844 = 180 + 764 \log(\text{days})$$

$$\text{days} = 7.3979 \dots$$

$$\approx 7.4 \text{ days}$$

b. What implication does this high dosage of drug have on the quality of life of these rats?

(1 mark)

The rats will keep increasing their markers and getting more unwell from the virus and possibly not recover.

16. The graph on page 16 of the High Dosage data could be considered a time series.

a. What would be the key feature/s of this plot?

(1 mark)

Positive Secular Trend

(1)

Structural

(1)

b. Show with a clearly marked 'X' on the graph on page 16, the seven-median smoothed point for day 8.

(1 mark)

c. Calculate the six-moving mean for day 8 to 2 decimal places.

(2 marks)

$$(855.83 \dots + 903.666 \dots) / 2 = 879.75$$

Covid-19 Confirmed Cases by Country as at 12/03/20

Country	Total Cases
China	80,796
Italy	12,462
Iran	9,000
S. Korea	7,869
France	2,281
Spain	2,277
Germany	1,966
USA	1,329
Diamond Princess	696
Switzerland	652
Japan	639
Norway	629
Denmark	514
Netherlands	503
Sweden	500
UK	456
Belgium	314
Qatar	262
Austria	246
Bahrain	195
Singapore	178
Malaysia	149
Australia	139
Hong Kong	130
Canada	118
48 countries	10-99
52 countries	1-9
TOTAL = 125	

log (cases)

4.

4.

3.

3.

3.

3.

3.

3.

2.

2.

2.

2.

2.

2.

2.

2.

2.

2.

2.

2.

2.

2.

2.

2.

2.

1.

0.

For this summary the cruise ship Diamond Princess is counted as a country.

17. What percent of these countries have confirmed cases over 1000? (1 mark)

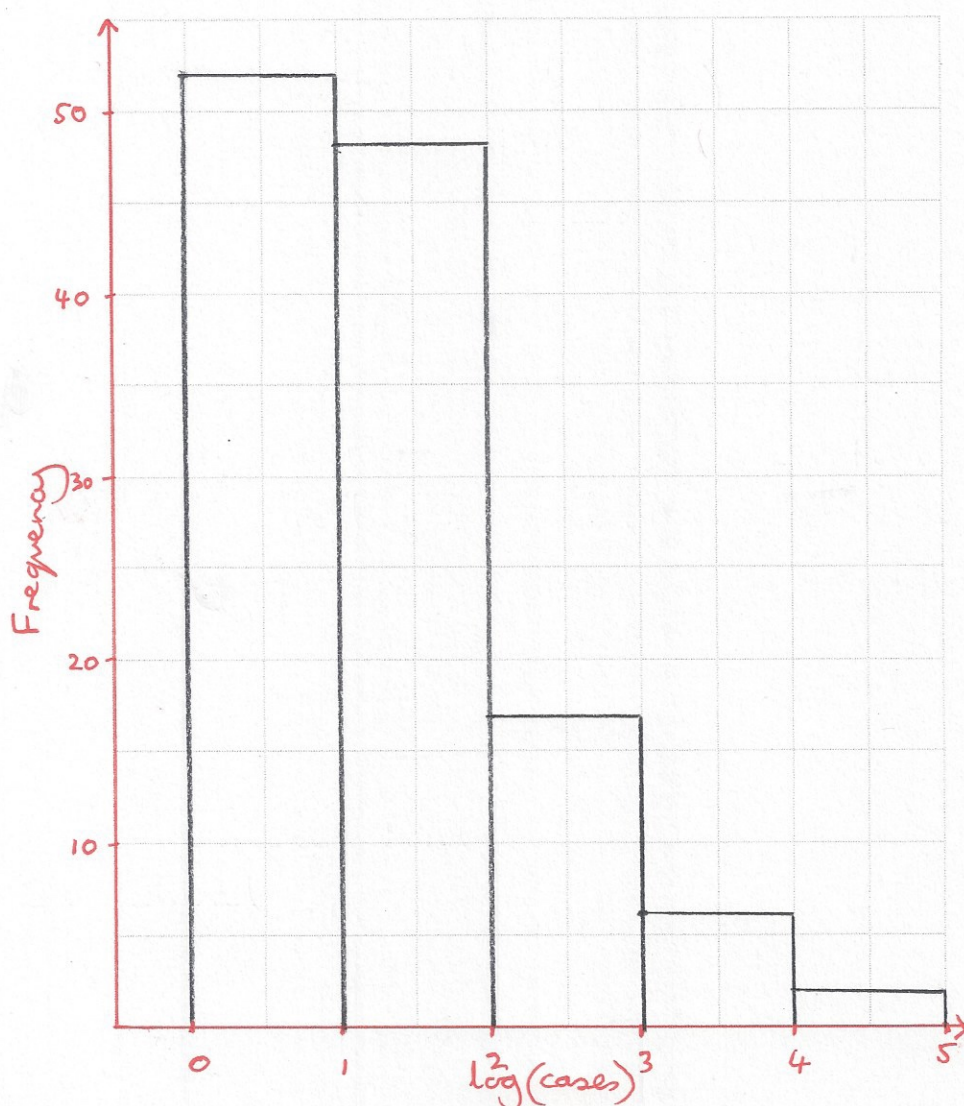
$$\frac{8}{125} \times 100 = 6.4\%$$

18. Using the table from the previous page, draw a histogram of the frequency against the $\log(\text{cases})$.

(3 marks)

$\log(\text{cases})$	Frequency
$4 < 5$	2
$3 < 4$	6
$2 < 3$	17
$1 < 2$	48
$0 < 1$	52

125



END OF SAC